

Madelyn Gatchel

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RESEARCH INTERESTS

I develop methods to study multi-agent strategic interactions using empirical game-theoretic analysis, focusing on learning game models from data, designing algorithms to analyze large games, and applying these techniques to mechanism design.

EDUCATION

University of Michigan

Ph.D. in Computer Science & Engineering

Ann Arbor, MI

September 2021 - Present

- Advisor: Professor Michael P. Wellman, Strategic Reasoning Group

M.S. in Computer Science & Engineering (GPA: 4.00)

August 2023

Davidson College

B.S. in Computer Science (High Honors) and Mathematics, Magna Cum Laude

Davidson, NC

May 2021

- GPA: 3.990 (CS: 4.000; Math: 3.970); Phi Beta Kappa, Omicron Delta Kappa; Thesis advisor: Dr. Bryce Wiedenbeck

AWARDS & HONORS

Rackham Outstanding Graduate Student Instructor Award

Spring 2025

- Recognized for outstanding instruction in EECS 592 AI Foundations (1 of 20 selected university-wide from ~2,000 GSIs).

2nd Place, CSE Graduate Student Honors Competition

Fall 2025

- Recognized for a research presentation in a competition featuring top doctoral students from five CSE research areas.

CSE HACKS Spirit Award

Spring 2024

- Awarded for exemplifying HACKS values: honesty, achievement, cooperation, knowledge, service (1 of ~600 students).

U-M Student Delegate, U7+ Alliance at NEXT Milan Forum

Winter 2024

- One of two U-M delegates; contributed to policy proposals on AI, sustainability, and global access to higher education.

PUBLICATIONS

1. **Madelyn Gatchel** and Michael P. Wellman. Learning Bayesian Game Families, with Application to Mechanism Design. In *25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2026.
2. **Madelyn Gatchel**. Game-Family Learning for Simulation-Based Games (Extended Abstract). In *24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS): Doctoral Consortium*, May 2025. 2929-2931.
3. **Madelyn Gatchel** and Bryce Wiedenbeck. Learning Parameterized Families of Games. In *22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2023. 1044-1052. **23% acceptance rate.**
4. **Madelyn Gatchel**. Analyzing Games with a Variable Number of Players (Extended Abstract). In *35th AAAI Conference on Artificial Intelligence: Student Papers and Demonstrations*, May 2021. Vol. 35, No. 18, 15960-15961.

PRIMARY INSTRUCTOR TEACHING EXPERIENCE

Division of Computer Science & Engineering, University of Michigan

Ann Arbor, MI

EECS 110 Discover Computer Science (31 students; 4.8/5.0 median eval)

Fall 2025

- Developed new lecture and lab materials for functions and libraries, reweighted assignment problems to better reflect course priorities, and standardized assignment instructions.
- Organized a student event on academic pathways in computer science beyond EECS 110.

EECS 110 Discover Computer Science (47 students; 4.8/5.0 median eval)

Fall 2024

- Built an assignment tracker, reorganized Canvas course, standardized rubrics, and prepared staff meeting materials.

ADDITIONAL TEACHING EXPERIENCE

Center for Research on Learning and Teaching in Engineering, University of Michigan

Ann Arbor, MI

Engineering Teaching Consultant

August 2025 - Present

- Co-facilitate the practice-teaching portion of undergraduate and graduate TA training (Fall 2025, Winter 2026).
- Conduct mid-semester classroom observations, solicit anonymous student feedback, and write reports for undergraduate and graduate engineering TAs.

Division of Computer Science & Engineering, University of Michigan

Ann Arbor, MI

Graduate Student Instructor

January 2024 - May 2024

- EECS 592 Artificial Intelligence Foundations (56 students; 4.7/5.0 median eval)

Department of Mathematics & Computer Science, Davidson College

Davidson, NC

Computer Science Assistant Teacher

August 2019 - May 2021

- CSC 121 Programming and Problem Solving (Python): Fall 2019, Spring 2020, Fall 2020 (60-90 students per semester)
 - Spring 2020: Taught three lectures (string manipulation, return statements, file I/O) during professor's absence.
 - Fall 2020: Updated quiz handouts, skeleton files, solutions, and autograders to align with restructured course
- CSC/MAT 220 Discrete Structures (introductory proofs course): Spring 2021 (25 students)

DEPARTMENT LEADERSHIP

CSE Ph.D. Admissions Committee, University of Michigan

Ann Arbor, MI

Graduate Student Representative

December 2021 - June 2024

Applicant Recruitment & Support (Fall):

- Created a statement-of-purpose feedback program pairing Ph.D. students with applicants from different backgrounds.
 - 2023: 16 applicants; 2024: 28 applicants, 8 acceptances, and 3 matriculations
- Represented UM CSE at Tapia Conference (2023) and UM CoE EMERGE program (2022).
- UM CSE Explore Graduate Studies: Panelist (2022, 2023) and Moderator (2023)

Admissions Committee Efforts:

- Provided input from a student perspective at weekly CSE Ph.D. admissions committee meetings.
- Designed a rubric system for current Ph.D. students to write thorough reviews of Ph.D. applications in AI (Dec 2022).
- Recruited and helped train ~50 total Ph.D. student volunteers to review Ph.D. applications in AI (Dec 2022, Dec 2023).
- Reviewed ~25 total AI-area applications for 2021-2022 and 2022-2023 application cycles.
- Evaluated ~20 AI-area applications for potential fellowship nomination during 2023-2024 application cycle.

Recruitment Efforts (Winter):

- Assisted the Admissions Committee faculty and the Graduate Programs Office in planning the virtual and in-person recruitment weekends, volunteering for most sessions of the visits (2022-2024).
- Recruited, scheduled, and managed ~50 Ph.D. student volunteers per year for virtual and in-person visits (2023, 2024).

CSE Division Chair Search Committee, University of Michigan

Ann Arbor, MI

Student Representative

July 2023 - December 2023

- Recommended strong prospective candidates, including one who advanced to the final interview stage.
- Conducted first- and second-round interviews with chair candidates and engaged in committee deliberations.
- Organized meetings for graduate students to meet with each second-round chair candidate.

AI Lab, University of Michigan

Ann Arbor, MI

AI Tea Co-Coordinator

December 2021 - May 2023

- Planned and facilitated monthly AI discussions with co-coordinator for 25+ AI lab graduate students and faculty.

Women in Computer Science (WiCS), Davidson College

Davidson, NC

Vice President

August 2019 - May 2021

- Co-organized weekly meetings and co-coordinated event logistics with outside speakers and groups.
- Mentored first-year and sophomore students exploring computer science.

MENTORING

- Steven Zhang, undergraduate research assistant
- Angelina Lee, instructional assistant for EECS 110 (Fall 2025)
- Ena Mestry, instructional assistant for EECS 110 (Fall 2024)

PROFESSIONAL SERVICE

- Workshop reviewer: GAIW-25, GAIW-26
- Conference sub-reviewer: AAMAS-26, ICAIF-25, NeurIPS-24, NeurIPS-23, EC-23
- Conference student volunteer: AAMAS-23, AAMAS-25

PROFESSIONAL DEVELOPMENT

- U-M Graduate Teacher Certificate (Spring 2026)
- Preparing Future Faculty Seminar (Summer 2025)
- AAMAS-25 Doctoral Consortium Scholar (Spring 2025)
- CRA-WP Grad Cohort for Women Scholar (Spring 2022)

VOLUNTEERING

- UM Provost Advisory Committee Meeting — Student Panelist (September 2023)
- CSE Undergraduate Research Symposium — Judge (April 2026)
- UM College of Engineering Director of Graduate Initiatives Search — Student Interviewer (April 2022)
- CSE Discussions — Presenter and Ph.D. Recruitment Discussion Moderator (October 2023)
- EECS 601: Introduction to CSE Graduate Research — Student Panelist (November 2022, November 2024)
- Women in Science and Engineering (WISE) Residential Program — Mentor (Fall 2023)
- Summer Research Opportunity Program (SROP) — Graduate Application Mentor (Summer 2023)
- UM Computing CARES Failing Forward — Student Panelist (February 2023)
- EECS 110 Discover Computer Science — Mentor (Winter 2022, Fall 2022)

UNDERGRADUATE HONORS & AWARDS

- AAAI-21 Undergraduate Consortium Scholar (Spring 2021)
- Davidson College Senior Computer Science Award (Spring 2021)
- Davidson College William Gillespie McGavock Mathematics Award (Spring 2021)
- Davidson College Bernard Society of Mathematics Sophomore Computer Science Award (Fall 2019)
- CRA-WP Grace Hopper Celebration Research Scholar (Fall 2019)
- DREU Undergraduate Research Assistant for Prof. R. Iris Bahar at Brown University (Summer 2019)
- Davidson College Chidsey Leadership Fellow (Spring 2018)